Project Title : Declining Sales in a Retail Business

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Exploratory Data Analysis Report

Topic : Analysis of Declining Sales in a Retail Business

1. Introduction

The purpose of this project is to clean retail sales data and perform Exploratory Data Analysis (EDA). EDA helps us understand the dataset, identify patterns, detect unusual observations, and summarize business performance.

This project focuses on sales revenue, profit, products, geographical regions, and sales channels. The analysis aims to investigate sales performance and identify areas that may require further attention.

2. Dataset Description

The dataset contains fictional retail sales records created for learning and analytical practice. The raw dataset contains 550 rows, including eight duplicate records. After removing duplicates, 542 unique order records remain.

The dataset contains the following fields:

- Order ID
- Order Date
- Product Name
- Quantity Sold
- Unit Price
- Discount
- Sales Revenue
- Cost
- Profit
- Region

- City
- Sales Channel

The dataset supports analysis of product performance, monthly sales, regional revenue, and sales channel performance.

3. Data Cleaning Process

The raw dataset was examined for missing values, duplicate records, incorrect formats, and inconsistent entries.

Missing values: Five Sales Revenue values, three Cost values, and three Profit values were missing. These values were recalculated using the documented synthetic dataset rules.

Duplicate records: Eight duplicate records were identified and removed, reducing the dataset from 550 to 542 unique orders.

Inconsistent entries: Product names, region names, and sales channel entries were standardized to maintain consistent spelling and capitalization.

Date formatting: Inconsistent date formats were converted to a consistent Excel date format.

Financial calculations: Sales Revenue was calculated as Quantity Sold multiplied by Unit Price and adjusted for Discount. Cost was calculated using the dataset's simplified 65% cost assumption, and Profit was calculated as Sales Revenue minus Cost.

These cleaning steps improved consistency and prepared the data for analysis.

4. Summary Statistics

The following statistics were calculated using Microsoft Excel formulas.

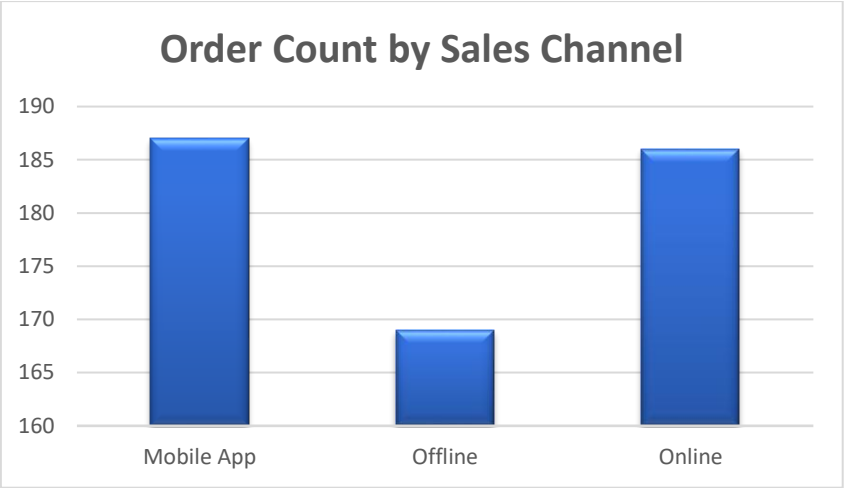
Metric	Result
Total Sales Revenue	₹1,79,34,579.88
Total Profit	₹50,51,802.51
Total Quantity Sold	2,411
Average Sales Revenue per Order	₹33,089.63
Total Unique Orders	542

The results provide an overview of sales volume, revenue, and profitability in the fictional dataset.

5. Exploratory Analysis and Charts

5.1 Order Count by Sales Channel

A column chart was created to compare orders across sales channels.

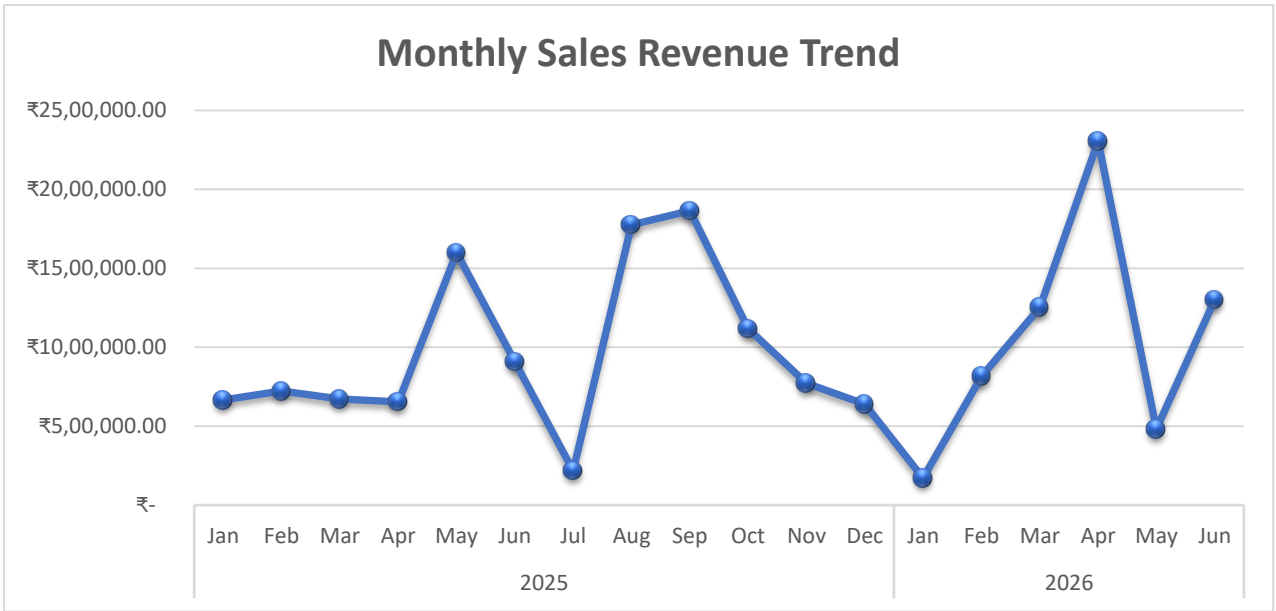


The Mobile App and Online channels have similar order counts, while the Offline channel has fewer orders.

5.2 Monthly Sales Revenue Trend

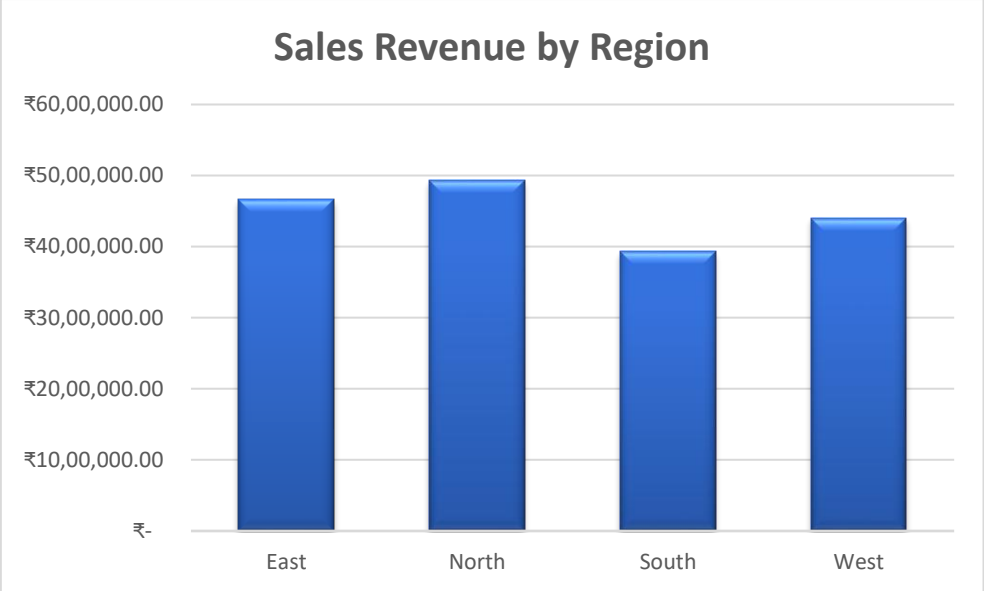
A line chart was created to examine changes in sales revenue over time. The monthly PivotTable was grouped by both year and month to show the chronological trend.

Monthly sales revenue fluctuates throughout the observed period. April 2026 recorded the highest monthly revenue of ₹23,06,290, while January 2026 recorded the lowest revenue of ₹1,71,466.38. July 2025 also showed unusually low revenue compared with the surrounding months. These periods require further investigation.



5.3 Sales Revenue by Region

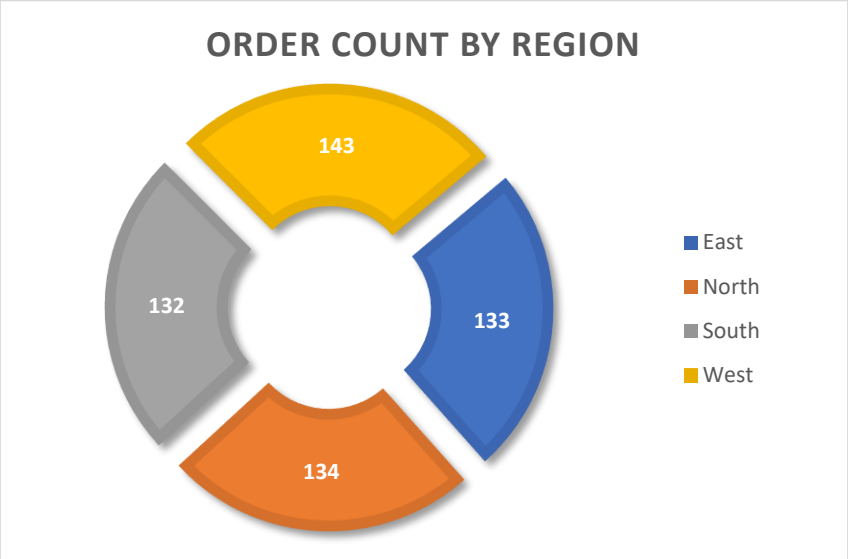
A column chart was created to compare regional revenue.



North generated the highest regional revenue, while South generated the lowest. Further investigation would be needed to understand the reasons for these differences.

5.4 Order Count by Region

A doughnut chart was created to show the distribution of orders across regions.



West had the highest order count, while South had the lowest. The regional order counts were relatively similar.

6. Key Observations

1. The cleaned dataset contains 542 unique orders.
2. Total sales revenue is approximately ₹1.79 crore.
3. North generated the highest revenue among the four regions.
4. The Mobile App and Online channels had similar order counts.
5. Monthly sales should be evaluated chronologically to determine whether the data shows a sustained decline or normal fluctuations.

These observations describe the supplied fictional dataset and do not establish the causes of sales performance.

7. Conclusion

This project demonstrates how data cleaning and exploratory analysis can support business decision-making. Missing financial values were handled using documented calculation rules, duplicate records were removed, and inconsistent entries and date formats were standardized.

Excel formulas, PivotTables, and charts were used to summarize sales revenue, profit, order counts, product performance, and regional performance.

The analysis identifies differences between regions and sales channels and provides a basis for investigating monthly sales patterns. Further analysis would be required to determine the causes of any sustained sales decline.

The dataset is fictional and intended for educational purposes. Its results should not be interpreted as actual business performance.